

Section B - Aptitude
10 questions | 35 minutes

- Q1. 4 mat-weavers can weave 4 mats in 4 days. At the same rate, how many mats would be woven by 8 mat-weavers in 8 days?
- Q2. How many seconds will a 500-meter-long train take to cross a man walking at a speed of 3 km/hr in the direction of the moving train if the speed of the train is 63 km/hr?
- Q3. Look at this series: 14, 28, 20, 40, 32, 64,... What number should come next?
- Q4. How many times do the hands of a clock coincide in a day?
- Q5. Today is Thursday. After 132 days, it will be
- Q6. Two dice are tossed. The probability that the total score is a prime number is:
- Q7. In how many ways can a committee, consisting of 5 men and 6 women can be formed from 8 men and 7 women?
- Q8. Mahendra Rajapaksa bought a total of 15 Aircraft, BMW, and Audi for ₹68 million. He bought more BMW than Audi. The cost of each Aircraft was ₹8 million each BMW was ₹3 million and each Audi was ₹4 million. How many Aircraft did he buy?
- Q9. The shopkeeper sold article A at a 20% loss and the cost price of article A was ₹4500. With that money, he bought article B and sold it at a profit of 30%. What is the overall profit or loss in the whole transaction?
- Q10. Excluding stoppages, the speed of a bus is 54 kmph, and including stoppages, it is 45 kmph. How many minutes does the bus stop per hour?





General Instructions:

- This question paper comprises 2 sections and a total of 20 questions.
- Section A comprises 10 programming questions for 40 minutes.
- Section B comprises 10 Aptitude questions for 35 minutes.
- There are no compilation errors, ignore missing things like, #include, [], etc.
- Incorrect answers will not result in Negative marks.
- Each question carries one mark.

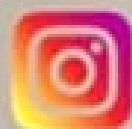
Section A - Programming
10 questions | 40 minutes

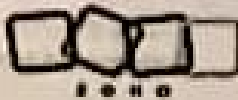
Q1. What is the output for the provided input in the program below?

```
1  Input: str1 = "aA", str2 = "aAAbbbb"
2
3  class Solution {
4      public int zoho(String str1, String str2) {
5          int num = 0;
6          for (int i = 0; i < str2.length(); i++) {
7              if(str1.indexOf(str2.charAt(i)) != -1)
8                  num++;
9          }
10         return num;
11     }
12 }
```

Q2. What is the output for the provided input in the program below?

```
1  Input: arr = {1,0,0,0,1}, n = 1
2
3  public class Solution {
4      public boolean zoho(int[] arr, int n) {
5          int count = 0;
6          for (int i = 0; i < arr.length; i++) {
7              if (arr[i] == 0) {
8                  boolean l = (i == 0) || (arr[i - 1] == 0);
9                  boolean r = (i == arr.length - 1) || (arr[i + 1] == 0);
10                 if (l && r) {
11                     arr[i] = 1;
12                     count++;
13                 }
14             }
15         }
```





```

16     return count == n;
17 }
18 }

```

Q3. What is the output for the provided input in the program below?

```

1  Input: strs = ["flower","flow","flight"]
2
3  public String zoho(String[] strs) {
4      if (strs.length == 0) return "";
5      String temp = strs[0];
6      for (int i = 1; i < strs.length; i++)
7          while (strs[i].indexOf(temp) != 0) {
8              temp = temp.substring(0, temp.length() - 1);
9              if (temp.isEmpty()) return "";
10         }
11     return temp;
12 }

```

Q4. What is the output for the provided input in the program below?

```

1  Input: n = 5
2
3  public int[] zoho(int num) {
4      int[] f = new int[num + 1];
5      for (int i = 1; i <= num; i++) f[i] = f[i > 1] + (i & 1);
6      return f;
7  }

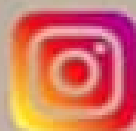
```

Q5. What is the output for the provided input in the program below?

```

1  Input: nums = [1,2,1]
2
3  class Solution {
4      public int[] zoho(int[] nums) {
5          int[] ans = new int[2 * nums.length];
6          for (int i = 0; i < nums.length; i++)
7              ans[i] = ans[i + nums.length] = nums[i];
8          return ans;
9      }
10 }

```





Q6. What is the output for the provided input in the program below?

```
1  Input: nums = [1,2,1,1,1,1]
2
3  class Solution {
4      public int zoho(int[] nums) {
5          int ans = 0;
6          for (int i = 0; i < nums.length; i++) {
7              for (int j = i + 1; j < nums.length; j++) {
8                  if (nums[i] == nums[j])
9                      ans++;
10             }
11         }
12         return ans;
13     }
14 }
```



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Q7. What is the output for the provided input in the program below?

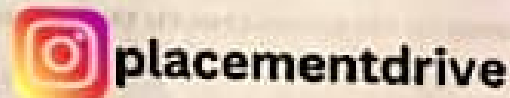
```
1  Input: nums = [2,2,3,1]
2
3  class Solution {
4      public int zoho(int[] nums) {
5          Pair<Integer, Boolean> a = new Pair<Integer, Boolean>(-1, false);
6          Pair<Integer, Boolean> b = new Pair<Integer, Boolean>(-1, false);
7          Pair<Integer, Boolean> c = new Pair<Integer, Boolean>(-1, false);
8          for (int num : nums) {
9              if ((a.getValue() && a.getKey() == num) ||
10                 (b.getValue() && b.getKey() == num) ||
11                 (c.getValue() && c.getKey() == num)) {
12                  continue;
13              }
14              if (!a.getValue() || a.getKey() <= num) {
15                  c = b;
16                  b = a;
17                  a = new Pair<Integer, Boolean>(num, true);
18              }
19              else if (!b.getValue() || b.getKey() <= num) {
20                  c = b;
21                  b = new Pair<Integer, Boolean>(num, true);
22              }
23              else if (!c.getValue() || c.getKey() <= num)
24                  c = new Pair<Integer, Boolean>(num, true);
25          }
26      }
27  }
```



```
26     if (tc.getValue())  
27         return a.getKey(i);  
28     return c.getKey(i);  
29 }  
30 }
```

Q8. What is the output for the provided input in the program below?

```
1  Input: nums1 = [2,4], nums2 = [1,2,3,4]  
2  public int[] solve(int[] nums1, int[] nums2) {  
3      int[] ans = new int[nums1.length];  
4      for(int i = 0; i < nums1.length; i++) {  
5          int temp = -1, j = nums2.length - 1;  
6          while(j >= 0 && nums2[j] != nums1[i]) {  
7              if(nums2[j] > nums1[i])  
8                  temp = nums2[j];  
9              j--;  
10         }  
11         ans[i] = temp;  
12     }  
13  
14     return ans;  
15 }
```



Q9. What is the output for the provided input in the program below?

```
1  Input: nums = [1,3,5,6], target = 7  
2  
3  public int zoho(int[] A, int target) {  
4      int low = 0, high = A.length-1;  
5      while(low < high){  
6          int mid = (low+high)/2;  
7          if(A[mid] == target) return mid;  
8          else if(A[mid] > target) high = mid-1;  
9          else low = mid+1;  
10     }  
11     return low;  
12 }
```

Q10. What is the output for the provided input in the program below?

```
1  Input: s = "abca"
```



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```
2
3 class Solution {
4     public boolean zoho(String s) {
5         int i = 0;
6         int j = s.length() - 1;
7         while(i <= j){
8             if(s.charAt(i) == s.charAt(j)){
9                 i++;
10                j--;
11            }
12            else return help(s, i + 1, j) ||
13        }
14        return true;
15    }
16    public boolean help(String s, int i, int j){
17        while(i <= j){
18            if(s.charAt(i) == s.charAt(j)){
19                i++;
20                j--;
21            }
22            else return false;
23        }
24        return true;
25    }
26 }
```